

MACHINEPULSE · REQUIREMENTS

MachinePulse requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/machinepulse		
Source	docs/03-requirements.md @ 3647c56		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	Status from MPL-CAL-001 for TRL 3; decisions from MPL-DDR-001 reflected; targets unchanged
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with concept status
0.1	2026-09-25	Amish Chadha	Initial scaffold

MachinePulse requirements

These are the MachinePulse requirements with their status from the TRL 3 calculation note MPL-CAL-001. Targets are unchanged from v0.2. The design they are checked against follows the recommendations adopted for TRL 3 in MPL-DDR-001 (Wi-Fi, USB adapter, IIS3DWB-class sensor, one CT, baseline alerts, TwinKit or any broker, stock box with stand-offs), which remain open for Amish's review. One requirement is not met (R10), six are at risk (R3, R4, R5, R8, R12 and R16), seven are met by calculation and three by design. How R4 should be treated with one CT is still awaiting Amish (MPL-DDR-001, O2).

Table 1. Requirements and TRL 3 status.

ID	Requirement	Target	Status (MPL-CAL-001)	Verification after TRL 3
R1	Detect run state	Off, idle and running classified from current, with 1 s time resolution	Met by calculation: 1 s RMS windows of 2,000 samples; zero-current reading 0.07 A against a 0.5 A threshold	Bench test on a motor
R2	Log run time	Run hours per machine, error within 1 % over a week	Met by calculation: worst case 1.00 % on 100 s runs, 0.17 % on 10 min runs	Comparison with a timed log
R3	Measure load current	Within 5 % of reading from 10 % to 100 % of CT range, after a two-point calibration	At risk: RSS 3.6 % at 10 % of range and 2.1 % at the design point, but 7.0 % worst case at 10 %; ESP32-S3 ADC residual assumed	Comparison with a clamp meter
R4	Estimate energy per shift	Within 10 % of a reference meter	At risk: 8.5 % RSS, 16.0 % worst case with a load-dependent power factor curve; 26 % with a fixed power factor. Treatment awaiting Amish (MPL-DDR-001, O2)	Comparison with a reference meter
R5	Measure vibration	Velocity RMS over 10 to 1,000 Hz (ISO 20816-1 band), three axes	At risk: sensor covers the band, but the magnet mount resonates at about 927 Hz on a painted curved frame (1,853 Hz flat), so readings are 10 % high from about 278 Hz	Mounted shaker check
R6	Low vibration noise floor	0.1 mm/s RMS or better, 10 to 1,000 Hz	Met by calculation: 0.037 mm/s (ADXL345-class fallback 0.20 mm/s would miss it)	Mounted noise test
R7	Resolve running-speed peaks	Spectrum bin 0.5 Hz or finer up to 1 kHz	Met by calculation: 0.407 Hz bins	Spectrum of a known tone
R8	Measure frame temperature	Within 2 °C of the surface, 0 to 100 °C	At risk: 1.56 K at 80 °C and 1.69 K at 85 °C, but about 3.6 K at 100 °C, where the sensor tolerance widens	Comparison with a thermocouple
R9	Report often enough	One summary per minute; spectrum every 15 min; delivered within 2 min	Met by calculation: 1.15 MB per day; 66 s worst delivery	Network log
R10	Install without opening live enclosures	Pod and probe in 10 min without tools; CT on an accessible insulated conductor	Not met for many machines. Pod and probe need no tools; the CT needs a single insulated conductor, often only inside a terminal box or panel opened by a qualified person	Design review with the pilot site
R11	Stay attached	No slip under 5 g peak on a painted cast frame; option for aluminium frames	Met by calculation: slip margin 3.3 on a curved painted frame (3.1 at 80 °C); 3 mm steel pads for aluminium frames, margin 7.5	Pull and shake tests
R12	Tolerate hot frames	Operate on frames up to 80 °C in 40 °C ambient	At risk: with 5 mm stand-offs the box floor reaches 53.9 °C and the module about 58.9 °C, but standard N-grade magnets reach their 80 °C limit on an 81.3 °C frame	Hot-plate test
R13	Workshop protection	IP54 or better for the pod	Met by design: IP54 box, IP68 glands, silicone boot round the sensor boss	IP test

ID	Requirement	Target	Status (MPL-CAL-001)	Verification after TRL 3
R14	Keep data local	Works with TwinKit or any MQTT broker on the local network; no cloud account	Met by design	Configuration check
R15	Store data when the network is down	7 days of summaries on the pod	Met by calculation: 13.9 days of summaries in 4 MB (4.0 days with spectra)	Offline run
R16	Low cost and buildable	Parts \$80 or less per machine; hand tools and a drill press; no custom PCB	At risk: \$80.00 against \$80, no margin, indicative prices	Supplier quotes
R17	Safe by design	Low voltage in the pod; certified 5 V adapter; voltage-output CT only; no connection to machine controls	Met by design; the CT input is now clamped against starting current	Design review

■ Assumptions

- Target machines are driven by induction motors from 0.37 kW to 15 kW, running at 50 or 60 Hz, at speeds from about 700 to 3,600 rpm.
- The CT is chosen per machine from the 5 to 60 A voltage-output range so that normal load sits above about 30 % of its range.
- A single CT on one phase is enough to judge run state and relative load on a three-phase motor whose phases are balanced; it is not an energy meter. Energy uses a power factor curve built from the nameplate and the measured no-load current.
- The pod is mounted on a clean flat or gently curved area of the frame or bearing housing, near the non-drive or drive-end bearing, with paint thickness under about 0.2 mm.
- The workshop has Wi-Fi coverage at the machine, or a TwinKit gateway nearby.